Dighinala Watershed

K-76

Figure 1:
Para (village) boundary
of Dighinala watershed

Disclaimer:
Boundaries are not necessarily authoritative

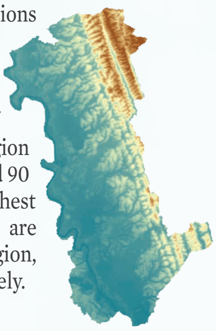
Source:
Center for Environmental andGeographic Information Services (CEGIS)

The Dighinala sub-watershed (K-76), located in the Dighinala, Babuchhara, Kabakhali Union of the Dighinala Upazila of the Khagrachhari District, encompasses 27 paras (figure 1) and covers approximately 3,991 hectares of land. It is home to almost 23,000 people, with communities consisting of the Manipuri, Marma, Tanchangya, and Tripura ethnic groups. The overall literacy rate is approximately 95%, yet opportunities for higher education are very limited. Farming is the main occupation. The average monthly income is 16,611 taka ($138), which is relatively low compared to the national average. About 78% of the houses are katcha and almost 24% of households use solar electricity systems. There is a noticeable scarcity of drinking water and a prevalent incidence of waterborne diseases.

Topography

Figure 2: Topography

The majority of the area (38%) is located at elevations below 50-meter (m) and an additional 33% at elevations between 50 and 70 meters. The valley is situated in the central to western region of the watershed. As elevation increases, the percentage of land area decreases significantly, with the eastern region covering 14% of the land at elevations between 70 and 90 meters and 6% between 90 and 110 meters. The highest elevations, ranging from 110 to over 130 meters, are predominantly found in the northeastern region, accounting for 5% and 4% of the total area, respectively.



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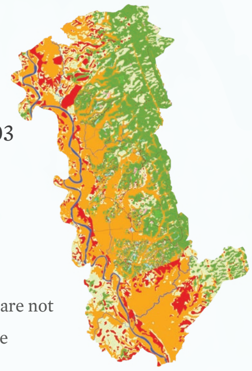
meter Mean Sea Level (mMSL)

Landuse

Figure 3: Land use

A total 12 types of land use have been identified within the K-76 watershed (figure 3). The main land use is forest (1,049 hectares (ha)), followed by herbaceous crop (982 ha), herb/shrub (803 ha), and crop land (549 ha).



- Forest Tree Dominated Area (26%)
- Crop Land (14%)
- Herb/Shrub Dominated Area (20%)
- Rural Settlement (11%)
- Shifting Cultivation (1%)
- Orchard (1%)
- Herbaceous Crop (25%)
- Others (3%)

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Demography

Population

51% 49%

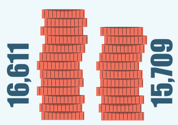
Total 13,277

Literacy

Male 96% **Female 95%**

Average literacy rate 95%

Economy



Monthly Income (BDT) Monthly Expense (BDT)

WASH

- 1667 tubewells (hand pumps)
- 5074 sanitary latrines
- 100 dug wells
- 28 chharas (creeks & Springs)

Poverty

Very Poor	22%
Poor	40%
Middle Class	32%
Better Off	6%

10 50 100

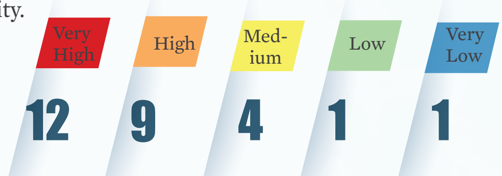
Production

- Rice 4-5 metric tons per hectare (MT/acre)
- Fisheries 28 -29 MT/acre
- 75,303 Livestock

Climatic Hazards

Climate Change-Induced Hazards

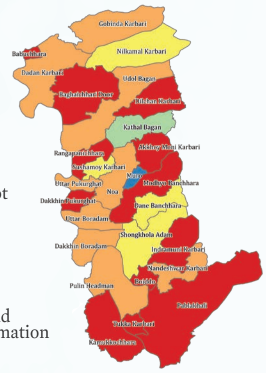
Nine climate hazards threaten Dighinala watershed agriculture, livelihood, ecosystem, and biodiversity. Flash floods and river floods intensify across almost all paras. Erratic rainfall, drought, heat waves, lightning, and landslides are increasing in frequency and severity, ranking highest in impact. Cold waves occur at low to moderate intensity.



Climate Change Risk

Figure 4: Mutihazard risk map

Climate change risk assessment identified 12 paras as very high-risk hotspots, nine as high-risk, four as medium-risk, one as low-risk, and the remaining para as very low-risk (Figure 4).



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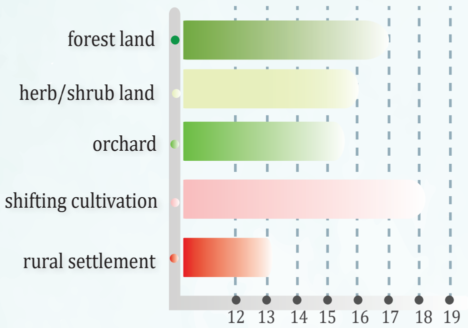
Watershed Degradation

Land Degradation Status

In K-76, land degradation is prominent in-



Causes of Degradation

- Jhum cultivation with short rotation
- Improper crop management
- Overgrazing
- Deforestation & overharvesting of trees
- Lack of vegetation cover
- Climate change
- Overuse of inorganic fertilizers

Degradation Triggering Factors

- Reduction of vegetation cover
- Quality and species composition change/diversity decline/monoculture plantation
- Soil erosion/landslides
- Surface sealing/hard pan development
- Waterlogging
- River bank erosion

Watershed Conservation

Conservation measures focused on enhancing ecological resilience as well as improving livelihood quality of the local community were identified and categorized into 6 classes.

Forest/ Shrub Land Conservation

20 ha
Forest enrichment plantation

3 kilometer (km)
Tree plantation along village road

Degraded Soil & Land Improvement

5
Landslide Treatment

15
Gully Treatment

Agriculture Land Conservation

40 ha
Jhum to agroforestry

80 ha
Fruit tree plantation

150 ha
Orchard management

3
Drainage improvement

Stream Bank Protection

3 km
Buffer strip plantation and bio-engineering methods

Water Resources Development

3
Irrigation canal improvement

5
Spring development

Capacity Development & Demonstration

56 Households (HHs)
Regenerative farming training

28 HHs
Drip irrigation training

28 HHs
Homestead improvement

Watershed Management Infrastructures

Culvert Construction	2
Earthen Dam Construction	29
Dam Improvement	13
Irrigation Canal Construction	40
Protection Wall Construction	0

Watershed conservation interventions in Dighinala include dam construction (29) and irrigation canal construction (40) being the most implemented, followed by dam improvement (13), aimed at enhancing water storage and distribution. In addition to these, other conservation measures have been identified to further boost water availability, agricultural productivity, and overall ecosystem health. Measures such as check dam construction, culvert construction, and protection wall construction will help control runoff, reduce soil erosion, and protect critical infrastructure. Together, these interventions will significantly benefit the local community by ensuring year-round water access, improving crop yields, and reducing vulnerability to climate-induced hazards. These integrated efforts promote sustainable livelihoods and strengthen resilience in the Dighinala region.

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